Task Report: Fashion Item Classification using Deep Learning (Fashion-MNIST)

1. Task Title

Fashion Item Classification using Deep Learning using TensorFlow and Keras (Without CNN)

2. Objective

Build a deep neural network using a Sequential model with Dense layers to classify fashion products (T-shirt, shoe, bag, etc.) from images using the Fashion-MNIST dataset, while applying core deep learning concepts such as activation functions, compilation, training, overfitting prevention and dropout.

3. Tools and Technologies

Programming Language: Python

Libraries: TensorFlow, Keras, NumPy, Matplotlib, Scikit-learn

Environment: Jupyter Notebook / Google Colab / VS Code

4. Dataset Information

Dataset: Fashion-MNIST

Source: https://github.com/zalandoresearch/fashion-mnist

Also directly available in Keras datasets.

Dataset contains:

60,000 training images

10,000 test images

28x28 grayscale images

10 classes:

0 T-shirt/top

1 Trouser

2 Pullover

3 Dress

4 Coat

5 Sandal

6 Shirt

7 Sneaker

8 Bag

9 Ankle boot

5. Dataset Loading

from tensorflow.keras.datasets import fashion_mnist

6. Data Preprocessing Steps

Step 1 Normalize pixel values:

Step 2 One-hot encode labels

7. Model Architecture

Sequential model

8. Model Compilation

9. Training the Model

10. Concepts Covered

Sequential Model: Linear stack of layers

Dense Layers: Fully connected layers

Activation Functions: ReLU and Softmax

Compilation: Loss, optimizer, metrics

Training: fit(), epochs, batch size

Overfitting: monitored using validation accuracy

Dropout: used for regularization

12. Model Evaluation

13. Sample Output

Input image: Sneaker

Predicted Output: Sneaker

14. Overfitting Analysis

15. Accuracy Graphs to Include in Report

16. Expected Output Screenshots

Include:

Dataset sample images

Model summary

Training logs

Accuracy graph

Prediction examples

17. Applications

Online fashion recommendation

Retail automation

Smart shopping assistants

Image classification learning

21. Conclusion

This project demonstrates how a feedforward neural network using TensorFlow/Keras can classify fashion images and apply fundamental deep learning concepts including Sequential models, Dense layers, training, dropout and overfitting control.